September 24, 2026

Dear Editor,

I am submitting “Model-Agnostic Interpretation of Machine Learning via Average Marginal Effects” for consideration at the Annals of Statistics, together with a supplement containing the proofs.

An earlier version of this work was declined by the Annals. The decision cited three concerns: “the novelty of the methodology, the lack of optimality results for the estimation problem, and the crucial root-n consistency condition, which is often not valid when the complexity of the function class grows.” I am grateful for those comments. A decision letter that identifies specific technical shortcomings is more useful than most referee reports, and the three named here were the right three: the paper as it stood had no optimality theory, and its rate condition was indeed one that flexible learners cannot meet. Rather than defend the earlier version I rebuilt it around those objections, and I am resubmitting because the result is a substantially different manuscript in which each point is addressed directly. I set out below how, so that the changes can be assessed quickly.

On the root-n condition, which was the substantive objection: the revision removes the condition rather than defending it. The estimand is now the window average marginal effect, the average centered difference of the regression function at a fixed step, rather than the expectation of a partial derivative. Because that functional is linear in the regression function, a change of variables yields an exact Riesz representation, with no boundary terms and no density derivatives anywhere in the argument. The estimation bias of the resulting orthogonal score is exactly a product of two L^2 errors, an identity rather than an approximation, with no linearization remainder to control. Root-n inference for the cross-fitted estimator therefore requires only that the product of the two nuisance L^2 errors be of smaller order than the parametric rate — for instance, both converging faster than the fourth root of n — rather than parametric-rate convergence of the learner in a function-space norm. This is precisely the condition that growing-complexity learners can satisfy. A double robustness result extends validity further: where the representer is known or parametrically estimable, inference is valid with no consistency requirement on the learner at all, which covers fixed-complexity learners converging to pseudo-true limits.

On optimality, the paper now establishes that the cross-fitted debiased estimator attains the semiparametric efficiency bound for the estimand: among all regular estimators, none has smaller asymptotic variance.

On novelty, three results are new. The first is the exact product-bias identity, which is what makes the rate condition as weak as it is. The second is the change-of-variables Riesz representation, which avoids the boundary terms and density derivative estimation that the average derivative literature has had to handle since Powell, Stock and Stoker. The third is a

pairing identity characterizing what the marginal effect of a piecewise constant learner measures, namely the jump measure of the fitted surface integrated against a locally averaged covariate density; this appears to be the first characterization of the population quantity tracked by the marginal effect of a tree-based model. Gradient consistency is also established for neural networks in Sobolev spaces and for gradient boosted models and Mondrian forests in the space of functions of bounded variation, delineating when the uncorrected plug-in is already consistent, with the obstruction for Breiman forests identified.

The empirical work now spans three datasets and three simulation studies. The credit default application is the sharpest illustration. Under the plug-in, four learners put the effect of the payment ratio at −0.431, −0.440, −0.045 and 0.0009, estimates that cannot all describe the same population. Under the correction they become −0.489, −0.584, −0.265 and −0.194. The disagreement was regularization bias, which differs across learners because their regularization differs, and which the orthogonal score removes.

A condensed version of this work has been accepted to the research track of IEEE ICDM 2026 and will appear in the conference proceedings. That version presents the plug-in estimator with a model-refitting bootstrap and reports the resulting coverage failure as an open problem. The overlap with the present manuscript is confined to the finite-difference construction and two of the three empirical applications; none of the theory appears there — not the window estimand, the Riesz representation, the orthogonal score, the efficiency result, the gradient consistency theorems, or the debiased estimator that resolves the failure the conference version could only report. I mention it in the interest of full disclosure.

The manuscript is my sole-authored work, is not under consideration elsewhere, and the accompanying software is publicly available.

Thank you again for the earlier comments, and for your consideration of this version.

Sincerely,

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